

7 Endocrine Surgery in Otolaryngology

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7.3 Thyroid Nodules and Cysts

◆ Key Features

- A thyroid nodule is a discrete lesion within the thyroid gland that is radiologically distinct from the surrounding thyroid parenchyma.
- Nodular disease of the thyroid gland is estimated to be present in 5 to 10% of the population.
- Thyroid nodules are more common in women.
- Less than 5% of thyroid nodules are malignant.

The prevalence of thyroid nodules is 2 to 6% by palpation, 19 to 35% with ultrasound, and 8 to 65% in autopsy data. The important question when a thyroid nodule is detected is whether the nodule is in fact malignant.

◆ Epidemiology of Nodules

Females are affected more than males. Nodule incidence increases with age and is increased in people with iodine deficiency and after radiation exposure. The prevalence of cancer is in thyroid nodules is higher in children, adults < 30 years or > 60 years old, patients with a history of head and neck radiation, and those with a family history of thyroid cancer. The prevalence of cancer may be lower in patients with multinodular goiters and autonomously hyperfunctioning ("hot") nodules.

◆ Clinical

The usual presentations of thyroid nodules are as a lump in the neck noted by the patient, as an incidental finding during routine physical examination, or during a radiologic procedure such as carotid ultrasonography or neck computed tomography (CT). Symptoms associated with increased risk of malignancy include a rapid increase in size, dysphagia, hoarseness (nerve involvement), and pain.

◆ Evaluation of Nodules

History

Pertinent history includes prior head and neck irradiation, thyroid carcinoma in a first-degree relative, rapid growth of the nodule, hoarseness, dysphagia, neck pain or pressure, and any symptoms of hyperthyroidism (e.g., due to an autonomous hot nodule) or hypothyroidism (e.g., due to Hashimoto's thyroiditis). Family history of pheochromocytoma, hyperparathyroidism, chronic constipation and diarrhea, hypertension may be indicative of a multiple endocrine neoplasia (MEN) syndrome.

Physical Exam

A complete physical exam focusing on the thyroid gland and adjacent cervical lymph nodes should be performed. The nodule should be palpated to note size, consistency, mobility, and tenderness. Vocal fold mobility (nasopharyngoscope) should be noted. Cancer may be suspected when nodules are > 4 cm in size, when they are fixed to the surrounding structures, or when there is associated lymphadenopathy or vocal fold paralysis.

Labs

Thyroid function tests should be obtained as part of the initial evaluation of a solitary thyroid nodule. Serum thyroglobulin (Tg) levels are not indicated in the evaluation of solitary thyroid nodule.

Imaging

- Nuclear imaging cannot reliably distinguish between benign and malignant nodules and is not required if nodules are present. However, in patients with a suppressed thyroid-stimulating hormone (TSH) level, a thyroid scan determines regional uptake or function and can be used as a secondary study.
- Diagnostic thyroid/neck ultrasound should be performed in all patients with a suspected thyroid nodule, nodular goiter, or radiographic abnormality suggesting a thyroid nodule incidentally detected on another imaging study.
- Diagnostic thyroid ultrasound will help to determine whether there is truly a thyroid nodule that corresponds to the palpable abnormality or a thyroid cyst; to look for other nonpalpable thyroid nodules that may also need to be biopsied and the location and the number of the nodules; and to guide fine-needle aspiration biopsy (FNAB) (**Table 7.4; Table 7.5; Table 7.6**).

Table 7.4 Ultrasound patterns of thyroid nodule and malignancy

High suspicion [malignancy risk 70–90%]:
Solid hypoechoic nodule or solid hypoechoic component of a partially cystic nodule with one or more of the following features: irregular margins, microcalcifications, taller than wide shape
Intermediate suspicion [malignancy risk 10–20%]:
Hypoechoic solid nodule without high-suspicion features
Low suspicion [malignancy risk 5–10%]:
Isoechoic or hyperechoic solid nodule, or partially (> 50%) cystic nodule, with eccentric solid area without high-suspicion features
Very low suspicion [< 3%]:
Spongiform or partially cystic nodules without high- or intermediate-suspicion features
Benign [< 1%]: Purely cystic nodules

Table 7.5 Sonographic features of thyroid nodule that warrant biopsy

Threshold for FNAB	
Solid nodule	
• With suspicious sonographic features	≥ 1.0 cm
• Without suspicious sonographic features	≤ 1.5 cm
Mixed cystic–solid nodule	
• With suspicious sonographic features	≥ 1.5–2.0 cm
• Without suspicious sonographic features	≤ 2.0 cm
Spongiform nodule	≥ 2.0 cm
Simple cyst	Not indicated
Suspicious cervical lymph node	FNAB node ± FNAB associated thyroid nodule(s)

Table 7.6 Bethesda Diagnostic Categories for thyroid cysts and nodules

Bethesda Diagnostic Category			Risk of malignancy	Usual management
I	Nondiagnostic or unsatisfactory	Cyst fluid only Virtually acellular specimen Other (obscuring blood, clotting artifact, etc.)	1 to 4%	Repeat FNAB with ultrasound guidance
II	Benign	Consistent with a benign follicular nodule (includes adenomatoid nodule, colloid nodule, etc.) Consistent with lymphocytic (Hashimoto's) thyroiditis in the proper clinical context Consistent with granulomatous (subacute) thyroiditis Other	0 to 3%	Clinical follow-up
III	Atypia of undetermined significance or follicular lesion of undetermined significance		5 to 15%	Repeat FNAB Molecular testing

(continued)

Table 7.6 Bethesda Diagnostic Categories for thyroid cysts and nodules (*Continued*)

IV	Follicular neoplasm of suspicious for a follicular neoplasm	Specify if Hurthle cell (oncocytic) type	15 to 30%	Molecular testing Surgical lobectomy
V	Suspicious for malignancy	Suspicious for papillary carcinoma Suspicious for medullary carcinoma Suspicious for metastatic carcinoma Suspicious for lymphoma Other	60 to 75%	Molecular testing Surgical lobectomy or Near-total thyroidectomy
VI	Malignant	Papillary thyroid carcinoma Poorly differentiated carcinoma Medullary thyroid carcinoma Undifferentiated (anaplastic) carcinoma Squamous cell carcinoma Carcinoma with mixed features (specify) Metastatic carcinoma	97 to 99%	Near-total thyroidectomy

- Thyroid sonography with survey of the cervical lymph nodes should be performed in all patients with known or suspected thyroid nodules.
- If serum TSH is low, a radionuclide thyroid scan may be used determine whether a nodule is “hot,” “warm,” or “cold”; a hot functioning nodule rarely harbors malignancy. Radionuclide scanning alone is not the most accurate technique to distinguish benign from malignant thyroid disorders.
- CT and magnetic resonance imaging (MRI) are used to evaluate adenopathy, multinodular goiters causing pressure symptoms, substernal goiter, airway and vascular displacement, and invasion of tumor. These studies, however, are not routinely employed.
- Consider a chest CT to evaluate displacement of trachea seen on plain chest radiograph.
- Focal ¹⁸F-fludeoxyglucose (FDG)–positron emission tomography (PET) uptake within a sonographically confirmed thyroid nodule conveys an increased risk of thyroid cancer, and FNAB is recommended for those nodules > 1 cm.

Other Tests

Generally, only nodules > 1 cm should be further evaluated, since they have a greater potential to be clinically significant cancers. Occasionally, there may be nodules < 1 cm that require evaluation because of suspicious findings or history.

FNAB is the single most important procedure for differentiating benign from malignant thyroid nodules.

Other cytopathologic yield of FNAB includes uniform follicular epithelium and abundant colloid (suggesting nodular or adenomatous goiters), inflammatory cells (suggesting thyroiditis), papillary cells (psammoma bodies), giant cells (suggesting papillary carcinoma), amyloid deposits stained with Congo red (suggesting medullary carcinoma), and undifferentiated cells (suggesting anaplastic carcinoma).

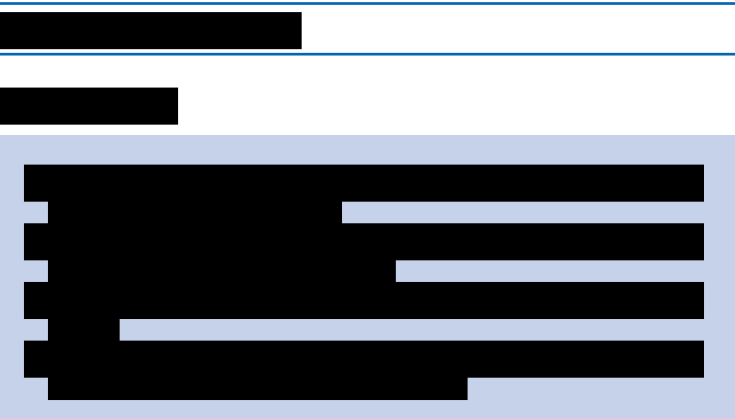
Serum TSH is an initial screening test and should be checked in all patients with thyroid nodules. If TSH is low, a radionuclide thyroid scan should be obtained to determine whether the nodule is functioning, or hot (uptake is greater in the nodule than in the surrounding normal thyroid). Functioning nodules are rarely malignant and so do not need to be evaluated further with FNAB. Further evaluation and treatment of the hyperthyroidism may be indicated. If the TSH is normal or high, a thyroid scan is of little value and should not be done.

Serum Tg is not routinely recommended, since it can be elevated in most thyroid diseases and is an insensitive and nonspecific test for thyroid cancer (it is the most valuable test as a marker of recurrence in follow-up of thyroid cancer after surgical treatment followed by radioactive iodine [RAI] treatment; see **Chapter 7.8**).

If thyroiditis is suspected, free thyroxine (FT₄), triiodothyronine (T₃), erythrocyte sedimentation rate (ESR), anti-thyroid peroxidase (anti-TPO), and anti-Tg should be checked only if they are clinically indicated. They have no routine role.

♦ **Thyroid Cysts**

Most cystic nodules of the thyroid are degenerating benign adenomas. Cysts that have a partial solid component are more suggestive of malignancy.



7.8 Thyroid Cancer

◆ Key Features

- Well-differentiated tumors (papillary and follicular) are highly treatable.
- Undifferentiated tumors (medullary or anaplastic) are aggressive and have a poorer prognosis.
- Thyroid cancer affects women more than men.
- Solitary nodules are more likely to be malignant in the young and the elderly.

Most thyroid malignancies are well-differentiated cancers that originate from follicular cells (papillary and follicular carcinomas). Thyroid tumors can also originate from the other cell types in the thyroid gland, including the calcitonin-producing C cells (parafollicular cells), lymphocytes, other vascular components, and metastases from other organs (**Table 7.8**).

Table 7.8 Overview of thyroid cancers

Thyroid cancer	Type	Incidence	Cell origin	Mutations	Subtype	Treatment
Differentiated	Papillary carcinoma	80%	Endodermally derived follicular cell	<i>BRAF</i> 45% <i>RET/PTC</i> 20% <i>RAS</i> 10% <i>TRK</i> < 5%		Surgery and RAI
	Follicular carcinoma	10%	Endodermally derived follicular cell	<i>RAS</i> 45% <i>PAX8-PPARγ</i> 35% <i>PIK3CA</i> < 10% <i>PTEN</i> < 10%	Hürthle cell	Surgery and RAI
Undifferentiated	Medullary carcinoma	5–10%	Neuroendocrine-derived calcitonin-producing C cell	Familial forms <i>RET</i> > 95% Sporadic <i>RET</i> 50%		Surgery

(continued)

Table 7.8 Overview of thyroid cancers (*Continued*)

	Anaplastic carcinoma	1–2%	Endodermally derived follicular cell	<i>TP53</i> 70% <i>β-catenin (CTNNB1)</i> 65% <i>RAS</i> 55% <i>BRAF</i> 20%	Small-cell carcinoma	Surgery when possible Chemotherapy Radiation therapy
					Giant-cell carcinoma	
Others	Lymphoma	Rare	Intrathyroid lymphoid tissue	<i>BRAF</i> <i>NRAS</i>	Diffuse large B-cell lymphoma	Chemotherapy Radiation therapy

Abbreviation: RAI, radioactive iodine.

◆ Epidemiology

Thyroid cancer represents the most common endocrine malignancy, annual incidence being ~ 64,000 cases in the United States with approximately 2,000 deaths. The female/male ratio is ~ 2:1 to 4:1. The incidence of this malignancy has been increasing over the last few decades. Annual incidence increases with age, peaking by the fifth through eighth decades. Thyroid cancer is very rare in children < 15 years. Patients with a history of radiation administered in childhood have an increased risk of cancer as well as other abnormalities of the thyroid gland.

◆ Well-Differentiated Thyroid Carcinomas

Well-differentiated thyroid carcinomas originate from the thyroid follicular cells and include papillary and follicular carcinomas, as described subsequently. These respond to thyroid-stimulating hormone (TSH), concentrate iodine, and synthesize thyroglobulin (Tg), albeit less efficiently than normal thyroid tissue. Well-differentiated thyroid carcinomas are two to four times more common in females than males.

Specifically, molecular analyses have focused on a set of somatic alterations of genes in the mitogen-activated protein kinase (MAPK) pathway that are frequently present in carcinomas of the thyroid. These include point mutations of the *BRAF* and *RAS* genes and *RET/PTC* and *PAX8/PPARγ* chromosomal rearrangements. A V600E mutation in the *BRAF* gene has been identified as the most common genetic event in papillary thyroid carcinoma, occurring in 40 to 45% of cases.

Papillary Thyroid Cancer

Papillary thyroid cancer (PTC) is the most common type of thyroid malignancy. It has a female/male ratio of 2.5:1. The tumor generally grows slowly, and the overall prognosis is excellent.

Pathogenesis

Mutations or rearrangements in the genes encoding for the proteins in the MAPK pathway such as *RET/PTC*, *RAS*, or *BRAF* have been implicated in the development and progression of differentiated thyroid cancer. It has been thought that the *BRAF* mutation is associated with more aggressive tumors with higher rates of extrathyroidal extension, lymph node metastases, and recurrence; however, this remains controversial. A major advance in recent years has been the discovery of mutations that have prognostic value, such as *TERT* (telomerase reverse transcriptase) promoter mutations.

Epidemiology

PTC accounts for 80 to 90% of all thyroid cancers. Its peak incidence is between ages 30 to 50 years. Females are more commonly affected than males.

Risk Factors

- History of radiation exposure during childhood, such as that received as a treatment of childhood malignancies, is a risk factor. Prior to about 1960, external radiation to the head and neck was commonly used to treat a wide variety of conditions, including enlarged tonsils or thymus and even acne.
- History of thyroid cancer in a first-degree relative or a family history of a thyroid cancer syndrome is a risk factor.
- Suggestive signs of a nodule's being malignant include a rapid increase in its size, its fixation to surrounding tissues, new-onset hoarseness or vocal fold paralysis, and the presence of ipsilateral cervical lymphadenopathy.

Clinical Presentation

PTC typically presents as a painless discrete mass in the thyroid gland. These tumors may be multicentric. The tumor generally grows slowly and is late to break through the capsule of the gland. Once it has become extrathyroidal, however, it may ultimately become invasive. Nodal metastases appear classically in paratracheal nodes but may be present anywhere in the neck. Bilateral spread is found in 8% of patients. It usually spreads via the lymphatic system. It is not uncommon to find microscopic foci of papillary carcinoma at autopsy or incidentally in a thyroid removed for other indications.

Pathology

Characteristic cytopathologic features seen on fine-needle aspiration biopsy (FNAB) or after surgical resection are diagnostic, including psammoma bodies (layered accumulations of calcium); nuclei with empty centers (Orphan Annie nuclei, based on the comic-strip character Little Orphan Annie) and the formation of papillary structures.

Prognosis

With treatment, overall outcome is generally favorable. However, a small group of patients develop local recurrence and/or distant metastases. Features associated with increased risk of recurrence or mortality include

age at diagnosis (age > 45 years), size of the primary tumor (> 2 cm), and the presence of soft tissue invasion and cervical lymph node or distant metastases.

Variants

PTC has several histologic subtypes, including a follicular variant, an oxyphilic variant, a solid or trabecular variant, and a clear cell variant. The follicular variant consists of several distinct subtypes. The diffuse follicular variant seems to present and behave in a more aggressive fashion.

Other biologically aggressive variants include a columnar variant, a tall cell variant, a diffuse sclerosing variant, and poorly differentiated carcinoma.

Follicular Thyroid Cancer

Epidemiology

Follicular thyroid cancer (FTC) is more common in the iodine-deficient regions of the world. It tends to occur in an older population, with a peak incidence between ages 40 and 60 years. Like most thyroid malignancies, FTC is more common in women (by about a 3:1 ratio).

Risk Factors

- Iodine deficiency may have a role in the pathogenesis, as there is a higher prevalence of FTC in iodine-deficient regions of the world, compared with iodine-sufficient regions.
- FTC is rarely associated with radiation exposure, *RET/PTC* mutations, or TSH receptor mutations.
- FTC has no association with familial syndromes.
- FTC may be associated with *RAS* mutations. *PAX8-PPARγ1* (a gene rearrangement) can be seen both in follicular adenomas and cancers.

Clinical Presentation

FTC typically presents as a painless solitary, mostly encapsulated nodule in the thyroid. FTC is more aggressive than PTC and usually spreads by hematogenous routes to bone or lung, and less commonly to brain and liver. It rarely invades lymphatics. The tumor is classified on the basis of degree of invasiveness into minimally invasive (encapsulated) or widely invasive. Metastases are more common with the widely invasive variant.

Diagnosis

FNAB cannot distinguish between follicular adenomas and carcinomas, because diagnosis of malignancy requires identification of tumor capsule and/or vascular invasion. Therefore, the actual diagnosis of follicular thyroid cancer is made on permanent pathologic evaluation of the thyroid specimen after surgery.

Prognosis

Factors associated with adverse prognosis in FTC include older age, distant metastases, large tumor size, vascular invasion, capsular extension,

histologic grade (widely invasive variant, Hürthle cell, insular and trabecular variants), and male sex.

Variants

Clear cell tumor, oxyphilic cell type or Hürthle cell type, and insular carcinoma are FTC variants.

Treatment of Well-Differentiated Thyroid Carcinomas

The primary treatment is surgical, followed by referral to an endocrinologist for medical management. Radioactive iodine remnant ablation treatment may be given if needed; there should be lifelong follow-up and surveillance for recurrence. Surgical treatments include:

- Selected papillary carcinomas that are < 1 cm in a young patient without a history of radiation exposure may be treated with hemithyroidectomy and isthmectomy followed by close observation. All others should be treated with a near total thyroidectomy and removal of any involved lymph nodes in the central or lateral neck areas. Elective lateral neck dissection is not recommended.
- FTC is treated with a total thyroidectomy.
- Hürthle cell carcinoma is treated with a total thyroidectomy and neck dissection in cases with clinically positive lymph nodes.

Postoperative Complications of Well-Differentiated Thyroid Carcinomas

Permanent hypoparathyroidism, transient hypoparathyroidism, damage to the recurrent laryngeal nerve (hoarseness), and damage to the superior laryngeal nerve are possible postoperative complications.

Postoperative Management of Well-Differentiated Thyroid Carcinomas

Radioactive Iodine Remnant Ablation

All patients with FTC and those PTC patients who have features associated with increased risk of recurrence or mortality (see the discussion of Prognosis under Papillary Thyroid Cancer) may undergo radioactive iodine (RAI) remnant ablation (RAI treatment or ¹³¹I remnant ablation). The RAI is taken up by the residual normal and tumor cells, leading to destruction or death of these cells. This not only reduces future recurrence risk but also facilitates surveillance for future recurrence. Before the treatment, plasma Tg is measured and a whole-body scan is performed after administration of a very small dose of RAI. Several days after the treatment, another whole-body scan is obtained (posttreatment scan).

Well-differentiated thyroid cancer has a reduced capacity to concentrate iodine compared with normal thyroid tissue. An elevated serum TSH stimulates the thyroid cancer cells to take up iodide enough to be detected by RAI imaging and synthesize and secrete Tg. TSH levels can be increased in two ways; one being withdrawal of thyroid hormone therapy. To achieve this, patients should be off of L-thyroxine (LT₄) for weeks prior to the whole-body

scan (because of longer half-life). Alternatively, patients may be started on liothyronine (Cytomel, Pfizer Inc., New York, NY; or LT_3) for 4 weeks after surgery, stopping 2 weeks before the whole-body scan. This has a shorter half-life than L-thyroxine, and TSH levels rapidly rise after its discontinuation. Alternatively, the TSH level can also be increased by administration of recombinant human TSH (RhTSH; Thyrogen, Sanofi Genzyme, Cambridge, MA).

TSH Suppression Therapy

After the diagnostic whole-body scan or treatment, the patient is started on L-thyroxine. The goal is to suppress TSH as much as possible. This serves two purposes: maintaining the patient in a euthyroid state and suppressing TSH and growth of any residual thyroid, as most well-differentiated tumors are TSH-responsive.

Long-Term Follow-Up and Surveillance for Recurrence

A TSH-stimulated whole-body scan is done at 6- or 12-month follow-up. The TSH stimulation is achieved by either thyroid hormone withdrawal or Thyrogen injection (depending on the individual case). Based on the results of this scanning, the decision is then made whether further treatment with RAI is needed.

Tg is a very useful marker for detecting the recurrence of thyroid cancer. Levels are measured periodically both while on thyroid hormone suppression and prior to whole-body scanning on TSH stimulation. A rising Tg level raises suspicion for recurrence, indicating the need for further treatment through RAI or revision neck surgery.

Imaging, such as ultrasound and positron emission tomography (PET)-computed tomography (CT) scanning, is important for detecting recurrences.

Patients need periodic clinical follow-up to monitor for symptoms such as hoarseness, hemoptysis, pain, dysphagia, cough and dyspnea, recurrent mass, new-onset adenopathy, or a paralyzed vocal fold.

Other Therapies for Well-Differentiated Thyroid Carcinomas

External beam radiation may play a role in the treatment of non-RAI-avid tumors, gross residual tumor, and unresectable disease, as well as in non-surgical candidates.

In general, thyroid cancers do not respond well to chemotherapy. A class of targeted drugs known as *kinase inhibitors* may help treat thyroid cancer cells with mutations in certain genes, such as *BRAF* and *RET/PTC*. Many of these drugs also affect tumor angiogenesis. Some kinase inhibitors that have shown early promise against thyroid cancer in clinical trials include sorafenib (Nexavar, Bayer, Leverkusen, Germany), sunitinib (Sutent, Pfizer Inc., New York, NY), and vandetanib (Caprelsa, AstraZeneca, Cambridge, UK). Other options include clinical trials involving gene therapy and tumor redifferentiation agents.

◆ Other Forms of Thyroid Cancer

Anaplastic Thyroid Cancer

Anaplastic thyroid cancer, a poorly differentiated cancer, accounts for ~ 2% of all thyroid cancers. Anaplastic thyroid cancer is the most aggressive and lethal of human neoplasms. Median survival is 6 months, despite treatment.

Epidemiology

The annual incidence of anaplastic thyroid cancer is 1 to 2 cases per million. The female/male ratio is 1.2 to 3.1:1. A higher incidence has been reported in areas of endemic goiter. Anaplastic thyroid cancer may arise de novo, but dedifferentiation from long-standing differentiated thyroid carcinoma is also suspected. Anaplastic thyroid cancer is responsible for > 50% of the deaths per year attributed to thyroid cancer. It is often a disease of the elderly, typically presenting in the sixth or seventh decade of life.

Clinical Presentation

It usually presents with local symptoms caused by a rapidly growing thyroid mass and extensive local invasion. Distant metastases may occur early in the course of the disease to the lungs, liver, bones, and brain.

Diagnosis

Diagnosis is suggested by cytologic examination of cells obtained by FNAB or core biopsy, although it is not always diagnostic, in which case surgical pathology is needed to confirm the diagnosis. Histopathologically, atypical cells are seen that show numerous mitoses and form different patterns. Multinucleate giant cells, spindle-shaped cells, and squamoid cells usually predominate.

Treatment

Surgical treatment may consist of complete resection in selected individuals when possible, followed by a combination of chemotherapy and radiotherapy. Palliative therapy includes a tracheotomy and chemoradiation. Often, the tumor is not resectable at presentation, and surgery consists of a tracheotomy or cricothyroidotomy to prevent airway compromise.

Radiation therapy is indicated preoperatively to increase the tumor resectability rate, or postoperatively to enhance the effect of chemotherapy or to alleviate obstruction, but its efficacy must be balanced against its toxicity.

Several chemotherapeutic agents have been used with uniformly suboptimal results. Doxorubicin has been shown to be useful in some patients. Tyrosine kinase inhibitors (such as imatinib [Gleevec, Novartis, Basel, Switzerland], sunitinib, or sorafenib) are under evaluation for the treatment of ATC.

Medullary Thyroid Cancer

Medullary thyroid cancer (MTC) is a neuroendocrine tumor of the C cells (parafollicular cells) of the thyroid gland. Calcitonin is secreted by the tumor

and is a useful marker for diagnosis and follow-up. MTC can be sporadic or familial.

Epidemiology

Seventy-five percent of MTCs are sporadic; the remainder are familial. There are three familial patterns of MTC: multiple endocrine neoplasia (MEN) 2A, MEN 2B, and familial MTC without other features of MEN. MTC is most aggressive in patients with MEN 2B.

Clinical Presentation

Medullary thyroid cancer typically presents as a painful hard nodule or mass in the thyroid gland or as an enlargement of the regional lymph nodes. Sometimes it comes to medical attention due to a metastatic lesion at a distant site.

Screening

RET proto-oncogene testing should be obtained in all patients with MTC. If the patient tests positive for the *RET* mutation, then genetic counseling and testing of family members should be offered. If screening is negative, no investigation of relatives is needed.

Pathology

Characteristic microscopic features are sheets of cells separated by a pink-staining substance that has characteristics of amyloid. Diagnosis can be confirmed by positive immunostaining of the tumor tissue for calcitonin and carcinoembryonic antigen. Preoperatively, patients should also be evaluated for hyperparathyroidism and for pheochromocytoma. Ultrasound or CT of the neck, a chest X-ray, and CT of the chest and upper abdomen should be obtained preoperatively to look for local and regional disease and distant metastases.

Treatment

Treatment is primarily surgical. Total thyroidectomy with removal of regional lymph nodes should be performed after excluding hyperparathyroidism and pheochromocytoma. In patients with MEN, surgery should be performed for pheochromocytoma prior to surgery for MTC. Patients with MEN often undergo prophylactic thyroidectomy in childhood to prevent the development of MTC.

Patients with MTC should receive a total thyroidectomy, a complete central neck dissection, and a lateral modified radical neck procedure on the side of the neck that harbored the tumor. Some recommend bilateral neck dissection. RAI is not a therapeutic option, because MTC does not take up iodine; therefore, it is imperative that surgery be thorough.

Postoperative Management

Patients should be started on thyroid hormone replacement (levothyroxine, LT_4) immediately after surgery. Because the C-cells are not stimulated by TSH, there is no role for TSH suppression.

Serum calcitonin and carcinoembryonic antigen (CEA) should be checked 6 months after surgery, to detect the presence of residual disease. An elevated basal serum calcitonin 6 or more months after surgery indicates residual disease.

Patients need continued follow-up with periodic physical examination and measurements of serum calcitonin and CEA. Patients with residual disease may benefit from radiotherapy. Vandetanib and cabozantinib (Cometriq, Exelixis, South San Francisco, CA) are tyrosine kinase inhibitors approved by the United States Food and Drug Administration (FDA) for progressive, metastatic medullary thyroid cancer. These agents target various tyrosine kinases, including MET, RET, and VEGFR-2.

Thyroid Lymphoma

Thyroid lymphomas are typical of the non-Hodgkin's lymphoma (NHL) type. Diffuse large-cell lymphoma is the most common type. Thyroid lymphoma may develop in the setting of longstanding Hashimoto's thyroiditis.

Epidemiology

Thyroid lymphoma represents approximately 2% of thyroid cancers. It is more common in women and usually presents in the sixth or seventh decade.

Clinical Presentation

Usually thyroid lymphoma presents as a rapidly enlarging goiter. Patients may experience symptoms or signs of compression of the trachea or esophagus, including dysphagia, dyspnea, stridor, hoarseness, and neck pain. On physical examination, the thyroid is usually firm, slightly tender, and is fixed to surrounding structures. It commonly extends substernally. It is not uncommon to see enlarged cervical or supraclavicular lymph nodes. Patients presenting with hoarseness usually have vocal fold paralysis.

In addition, 10% of patients have systemic (B) symptoms of lymphoma, including fever, night sweats, and weight loss (10% of body weight or more). Patients may also present with symptoms and signs of hypothyroidism or hyperthyroidism.

Diagnosis

An ultrasound of the thyroid gland should be performed. FNAB or core biopsy distinguishes thyroid proliferation from epithelial tumors and may differentiate lymphoma from chronic thyroiditis; often surgical specimens are required for diagnosis. Pathology, immunohistochemical staining, or flow cytometry may be necessary to establish monoclonality and characterize surface markers, especially to diagnose small cell lymphomas.

Staging includes physical examination, complete blood count (CBC), serum lactate dehydrogenase (LDH), B2 microglobulin measurements, liver function tests, bone marrow biopsy, and CT of the neck, thorax, abdomen, and pelvis, as well as an ^{18}F fludeoxyglucose (FDG)-PET-scan.

Treatment

Surgery is not the primary treatment and is typically used for diagnostic biopsy and surgical airway only. Treatment depends on extent of disease. If

lymphoma is disseminated, chemotherapy is given. If disease is confined to the neck, treatment is guided by the histologic features of the lymphoma. Patients with large-cell lymphoma are treated with chemotherapy with or without radiation. For patients with localized extranodal marginal zone lymphoma of the thyroid, follicular lymphoma, or small-cell lymphoma, radiotherapy alone may be adequate (**Fig. 7.3**).

◆ Staging of Thyroid Cancer

TNM and Stage Groupings

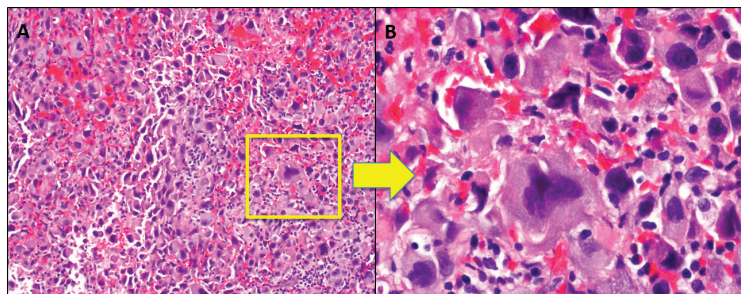


Fig. 7.3 Histology of anaplastic thyroid cancer (ATC). **(a)** Low-power view demonstrating abnormally shaped cells without papillary or follicular architecture. **(b)** High-power view demonstrating irregularly shaped cells and heterogeneity of nuclei, features typical of ATC. (Used with permission from Terris DJ, Duke WS, eds. *Thyroid and Parathyroid Diseases: Medical and Surgical Management*. 2nd ed. New York, NY: Thieme;2016:149.)

According to the American Joint Committee on Cancer 7th edition (AJCC), the TNM stage groupings for papillary and follicular carcinomas and variants are stratified by age into patients < 45 years of age and patients ≥ 45 years. In the 8th edition, effective January 2018, the age stratification is now < 55 years and ≥ 55 years. Tumor size and lymph node status are also considered in the TNM classification.

All categories may be subdivided into solitary tumor or multifocal tumor. With multifocal tumors, the largest one is used for classification. The lymph nodes must be specifically identified to classify regional node involvement.

Staging Thyroid Cancer Differentiated and Anaplastic (AJCC 8th edition)

Primary Tumor (T): Papillary, Follicular, Poorly Differentiated, Hürthle Cell, and Anaplastic Thyroid Carcinoma

TX: Primary tumor cannot be assessed

T0: No evidence of primary tumor

T1: Tumor ≤ 2 cm in greatest dimension limited to the thyroid

T1a: Tumor ≤ 1 cm, limited to the thyroid

T1b: Tumor > 1 cm but ≤ 2 cm in greatest dimension, limited to the thyroid

- T2: Tumor > 2 cm but $\leq$ 4 cm in greatest dimension, limited to the thyroid
- T3: Tumor > 4 cm limited to the thyroid or gross extrathyroidal extension invading only strap muscles
- T3a: Tumor > 4 cm limited to the thyroid
- T3b: Gross extrathyroidal extension invading only strap muscles (sternohyoid, sternothyroid, thyrohyoid, or omohyoid muscles) from a tumor of any size
- T4: Includes gross extrathyroidal extension
- T4a: Gross extrathyroidal extension invading subcutaneous soft tissues, larynx, trachea, esophagus, or recurrent laryngeal nerve from a tumor of any size
- T4b: Gross extrathyroidal extension invading prevertebral fascia or encasing the carotid artery or mediastinal vessels from a tumor of any size

Note: All categories may be subdivided: (S) solitary tumor or (M) multifocal tumor (the largest tumor determines the classification).

Regional Lymph Node (N)

- NX: Regional lymph nodes cannot be assessed
- N0: No evidence of locoregional lymph node metastasis
- N0a: One or more cytologically or histologically confirmed benign lymph nodes
- N0b: No radiologic or clinical evidence of locoregional lymph node metastasis
- N1: Metastases to regional nodes
- N1a: Metastases to level VI or VII (pretracheal, paratracheal, or prelaryngeal/delphian, or upper mediastinal) lymph nodes; this can be unilateral or bilateral disease
- N1b: Metastasis to unilateral, bilateral, or contralateral lateral neck lymph nodes (levels I, II, III, IV, or V) or retropharyngeal lymph nodes

Definition of Distant Metastasis (M)

- M0: No distant metastasis
- M1: Distant metastasis

American Joint Committee on Cancer Stage Groupings of Differentiated and Anaplastic Thyroid Cancer (8th Edition)

Differentiated: See **Table 7.9**. Anaplastic: See **Table 7.10**.

Histopathologic Type

- Papillary carcinoma
- Papillary microcarcinoma
 - Follicular variant
 - Solid variant

- Hürthle cell variant
- Follicular carcinoma
 - Encapsulated noninvasive
 - Minimally invasive
 - Widely invasive
- Hürthle cell carcinoma
- Poorly differentiated carcinoma (used for insular carcinoma as a subtype of poorly differentiated)
- Anaplastic carcinoma

Table 7.9 American Joint Committee on Cancer 8th edition stage groups, differentiated thyroid cancers

When age at diagnosis is...	and T is...	and N is...	and M is...	then the stage group is...
< 55 years	Any T	Any N	M0	I
< 55 years	Any T	Any N	M1	II
≥ 55 years	T1	N0/NX	M0	I
≥ 55 years	T1	N1	M0	II
≥ 55 years	T2	N0/NX	M0	I
≥ 55 years	T2	N1	M0	II
≥ 55 years	T3a/T3b	Any N	M0	II
≥ 55 years	T4a	Any N	M0	III
≥ 55 years	T4b	Any N	M0	IVA
≥ 55 years	Any T	Any N	M1	IVB

Used with permission from Amin MB, Edge S, Greene F, et al, eds. *AJCC Cancer Staging Manual* 8th Edition. New York: Springer; 2017.

Table 7.10 American Joint Committee on Cancer stage groups, anaplastic thyroid cancers

When T is...	and N is...	and M is...	then the stage group is...
T1–T3a	No/NX	M0	IVA
T1–T3a	N1	M0	IVB
T3b	Any N	M0	IVB
T4	Any N	M0	IVB
Any T	Any N	M1	IVC

Used with permission from Amin MB, Edge S, Greene F, et al, eds. *AJCC Cancer Staging Manual* 8th Edition. New York: Springer; 2017.

Medullary Thyroid Cancer*Primary Tumor (T)*

- TX: Primary tumor cannot be assessed
- T0: No evidence of primary tumor.
- T1: Tumor ≤ 2 cm in greatest dimension limited to the thyroid
 - T1a: Tumor ≤ 1 cm in greatest dimension limited to the thyroid
 - T1b: Tumor > 1 cm but ≤ 2 cm in greatest dimension limited to the thyroid
- T2: Tumor > 2 cm but < 4 cm in greatest dimension limited to the thyroid.
- T3: Tumor ≥ 4 cm or with extrathyroidal extension
 - T3a: Tumor ≥ 4 cm in greatest dimension limited to the thyroid
 - T3b: Tumor of any size with gross extrathyroidal extension invading only strap muscles (sternohyoid, sternothyroid, thyrohyoid, or omohyoid muscles)
- T4: Advanced disease
 - T4a: Moderately advanced disease; tumor of any size with gross extrathyroidal extension into the nearby soft tissues of the neck, including subcutaneous soft tissues, larynx, trachea, esophagus, or recurrent laryngeal nerve
 - T4b: Very advanced disease; tumor of any size with extension toward the spine or into nearby large blood vessels, invading prevertebral fascia or encasing the carotid artery or mediastinal vessels

Regional Lymph Node (N)

- NX: Regional lymph nodes cannot be assessed
- N0: No evidence of locoregional lymph node metastasis
 - N0a: One or more cytologically or histologically confirmed benign lymph nodes
 - N0b: No radiologic or clinical evidence of locoregional lymph node metastasis
- N1: Metastases to regional nodes
 - N1a: Metastases to level VI or VII (pretracheal, paratracheal, or prelaryngeal/delphian, or upper mediastinal) lymph nodes; this can be unilateral or bilateral disease
 - N1b: Metastasis to unilateral, bilateral, or contralateral lateral neck lymph nodes (levels I, II, III, IV, or V) or retropharyngeal lymph nodes

Distant Metastasis (M)

- M0: No distant metastasis
- M1: Distant metastasis

American Joint Committee on Cancer Stage Groupings of Medullary Thyroid Cancer (8th edition)

See **Table 7.11**.

Grade is not used in the staging for MTC.

When T is...	and N is...	and M is...	then the stage group is...
T1	N0	M0	I
T2	N0	M0	II
T3	N0	M0	II
T1–3	V1a	M0	III
T4a	Any T	M0	IVA
T1–3	N1b	M0	IVA
T4b	Any T	M0	IVB
Any T	Any T	M1	IVC

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[illegible]